

Repetitions to failure versus not to failure during concurrent training in healthy elderly men: A randomized clinical trial

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ABSTRACT

This randomized clinical trial compared the neuromuscular adaptations induced by concurrent training (CT) performed with repetitions to concentric failure and not to failure in elderly men. Fifty-two individuals (66.2 ± 5.2 years) completed the pre- and post-measurements and were divided into three groups: repetitions to failure (RFG, $n = 17$); repetitions not to failure (NFG, $n = 20$); and repetitions not to failure with total volume equalized to RFG (ENFG, $n = 15$). Participants were assessed in isometric knee extension peak torque (PT_{iso}), maximal strength (1RM) in the leg press (LP) and knee extension (KE) exercises, quadriceps femoris muscle thickness (QF MT), specific tension, rate of torque development (RTD) at 50, 100 and 250 ms, countermovement jump (CMJ) and squat jump (SJ) performance, as well as maximal neuromuscular activity (EMG_{max}) of the *vastus lateralis* (VL) and *rectus femoris* (RF) muscles. CT was performed over 12 weeks, twice weekly. Along with each specific strength training program, each group also underwent an endurance training in the same session. After training, all groups improved similarly and significantly in LP and KE 1RM, PT_{iso} , CMJ and SJ performance, RTD variables, specific tension, and VL EMG_{max} ($P < 0.05$ – 0.001). QF MT improved only in RFG and ENFG ($P < 0.01$). These results suggest that repetitions until concentric failure does not provide further neuromuscular performance gains and muscle hypertrophy, and that even a low number of repetitions relative to the maximal possible (i.e., 50%) optimizes neuromuscular performance in elderly men. Moreover, training volume appears to be more important for muscle hypertrophy than training using maximal repetitions.

1. Introduction

Biological aging is associated with declines in maximal strength, muscle mass and quality, muscle explosiveness (i.e., rate of force or torque development), muscle power output, as well as cardiorespiratory capacity, which results in an impaired capacity to perform activities of daily living (Fleg and Lakatta, 1998; Izquierdo et al., 1999a, 1999b; Aagaard et al., 2010). In view of this, concurrent strength and endurance training seems to be the best strategy to counteract this process in healthy elderly individuals, because it induces both neuromuscular and cardiovascular gains (Wood et al., 2001; Izquierdo et al., 2004; Cadore et al., 2010, 2011; Ferrari et al., 2013).

Performing repetitions until concentric failure has been widely used during strength training (ST) regimens (Kraemer et al., 1999; Cadore

et al., 2010; Izquierdo-Gabarrén et al., 2010; Sampson and Groeller, 2016). In healthy elderly people, ST using repetitions to failure seems to result in marked neuromuscular gains (Izquierdo et al., 2004; Kraemer et al., 1999; Pinto et al., 2014; Cadore et al., 2012). However, several studies have shown that ST performed with repetitions until concentric failure does not induce additional muscle strength and power output gains when compared to repetitions not to failure (i.e., submaximal repetitions per set) in young populations (Folland et al., 2002; Izquierdo et al., 2006; Izquierdo-Gabarrén et al., 2010; Sampson and Groeller, 2016; Martorelli et al., 2017), whereas a fewer number of studies observed greater strength gains following repetitions to failure (Rooney et al., 1994; Drinkwater et al., 2005). In addition, it seems that ST with repetitions to failure (i.e., maximal repetitions per set) does not induce further muscle size gains in young subjects (Sampson and

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Groeller, 2016; Martorelli et al., 2017; Nóbrega et al., 2018), although its effects are less investigated. Notwithstanding, to the best of the authors' knowledge, no previous study has compared the performance of ST with repetitions to failure or not to failure (i.e., submaximal per set) in elderly populations. Moreover, it is still unclear what is the minimal number of repetitions needed, in relation to the maximal possible, to optimize the neuromuscular adaptations in the elderly; this issue needs to be further investigated.

Although ST performed with repetitions until concentric failure induces improvements in several neuromuscular parameters (Cadore et al., 2010; Pinto et al., 2014; Cadore et al., 2013), it also results in a longer time under tension, which induces greater increases in the blood pressure, heart rate, and rate-pressure product (Nery et al., 2010; Lovell et al., 2011; Gjovaag et al., 2016), which could increase the cardiovascular risk in elderly. In addition, ST with repetitions to failure induces greater metabolic impact (Gorostiaga et al., 2012), which may result in greater time of recovery necessary between exercise sessions. Thus, it seems relevant to compare the effects of performing repetitions to failure and not to failure in the neuromuscular adaptations to training in elderly, giving special attention to the influence of total volume on these adaptations (i.e., compensating or not compensating the number of repetitions with additional sets). Moreover, this information would be especially useful in the context of concurrent training, since it is an effective strategy to improve overall neuromuscular and cardiovascular functions in elderly (Cadore and Izquierdo, 2013). Therefore, the aim of the present study was to compare the neuromuscular adaptations induced by three types of concurrent training interventions in healthy elderly men: one with the ST performed with repetitions until concentric failure; another with ST performed with 50% of the repetitions to concentric failure; and a third, with ST performed with 50% of the repetitions to concentric failure, but equalizing the total volume by adding more sets. Our hypothesis was that all training groups would induce similar neuromuscular performance gains, although we expected that the groups with greater ST volume would have greater muscle size gains.

2. Materials and methods

2.1. Experimental design

To investigate the effects of concurrent training composed by ST workout performed with repetitions until concentric failure in elderly individuals, three training groups performed 12 weeks of different interventions. Because we also aimed to isolate the effects of ST volume (i.e., sets $\times$ repetitions), two training groups performed submaximal repetitions (i.e., 50% of the possible maximal repetitions), but one of them compensated for the lower number of repetitions per set by performing double the number of sets than the group which performed repetitions to failure, which resulted in equal volume. To test the stability and reliability of the performance variables, a subsample of the participants were assessed twice before the start of training (weeks -4 and 0). Pre- and post-intervention testing was performed by the same investigator, who was blinded to the training group to which the participants belonged. Exception of blindness was in the 1 RM variables, in which assessors were not completely blinded regarding individuals' groups, but blinded in relation to the pre-training values. The ambient conditions were kept constant throughout all tests (temperature: 22–24 °C) and interventions. This randomized clinical trial (RCT) was conducted according to the Declaration of Helsinki and approved by the local Institutional Ethics Committee (register number 39550914.3.0000.5347).

2.2. Participants

The complete screening, recruitment, and allocation of individuals are presented in the “Results” section (Fig. 1). Fifty-two healthy

community-dwelling elderly men (mean $\pm$ SD: 66.2 $\pm$ 5.2 years) who had not engaged in any regular and systematic training program in the previous 3 months participated in this study after completing an ethical consent form. The participants volunteered for the present investigation following announcements in widely read local newspapers, social media, and announcements at a local University. The participants were carefully informed about the design of the study, and special information was given regarding the possible risks and discomfort related to the procedures. Subsequently, the participants were randomly assigned and placed into three groups: concurrent training with the ST performed using repetitions until concentric failure (RFG, $n = 17$); concurrent training with the ST performed using repetitions not to failure (NFG, $n = 20$); and, concurrent training with the ST performed using repetitions not to failure, but with equalized total ST volume, comparing to RFG (ENFG, $n = 15$). Concealment was guaranteed by a researcher who was blinded with respect to participants. Twelve individuals (age: 68.0 $\pm$ 5.2 years; body mass: 83.0 $\pm$ 9.7 kg; height: 170 $\pm$ 7 cm; body mass index: 28.7 $\pm$ 4.2 kg·m⁻²) were evaluated twice before the start of training to provide control period data (weeks -4 and 0).

Medical evaluations were performed using clinical anamnesis and an effort electrocardiograph (ECG) test to ensure each subject's suitability for the testing procedure. The exclusion criteria included any history of neuromuscular, metabolic, hormonal and cardiovascular diseases (except controlled stage 1 hypertension). In addition, exclusion criteria also included smoking or having stopped smoking less than one year prior to the participation in the study. The participants were not taking any medications that could influence hormonal or neuromuscular metabolism. The participants were advised to maintain their normal dietary intake throughout the study. The physical characteristics of the participants were assessed as described elsewhere (Cadore et al., 2013) and in the Supplementary file, as well as are shown in Table 1.

2.3. Maximal dynamic strength

Maximal strength was assessed using the one-repetition maximum test (1 RM) on the bilateral leg press (LP) and bilateral knee extension (KE) exercises (KonnenGym, Beijing, China). More details on familiarization, warm-up and procedures are described elsewhere (Cadore et al., 2013), and in the Supplementary file. Each subject's maximal load was determined with no more than five attempts with a five-minute recovery between attempts. Performance time for each phase (concentric and eccentric) was 2 sec, controlled by an electronic metronome (Quartz, CA, USA).

2.4. Isometric peak torque and rate of torque development

Maximal isometric peak torque (PT_{iso}) was obtained using an isokinetic dynamometer (Cybex Norm, New York, USA). The dynamometer was connected to a 2000 Hz A/D converter (Miotec, Porto Alegre, Brazil), which made it possible to quantify the torque exerted when each subject executed the knee extension at the determined angle. Participants were seated with their hips and thighs firmly strapped to the seat of the dynamometer, with the hip angle at 85° and the lateral femoral condyle of the right leg was aligned with the axis of rotation of the dynamometer. More details on warm-up as well as on assessment and analysis procedures are described elsewhere (Cadore et al., 2013), and in the Supplementary file. Three 5-second knee extensions were performed with 120° in the knee extension (180° represented the full extension), with 2 min of rest interval between each attempt. All participants were instructed and encouraged to exert maximum strength “as hard and as fast as possible” after the starting command. Signal processing included filtering at a cut-off frequency of 10 Hz. Maximal peak torque was defined as the highest value of the torque (N·m) recorded during the unilateral knee extension. The isometric torque-time analysis on the absolute scale included the maximal torque

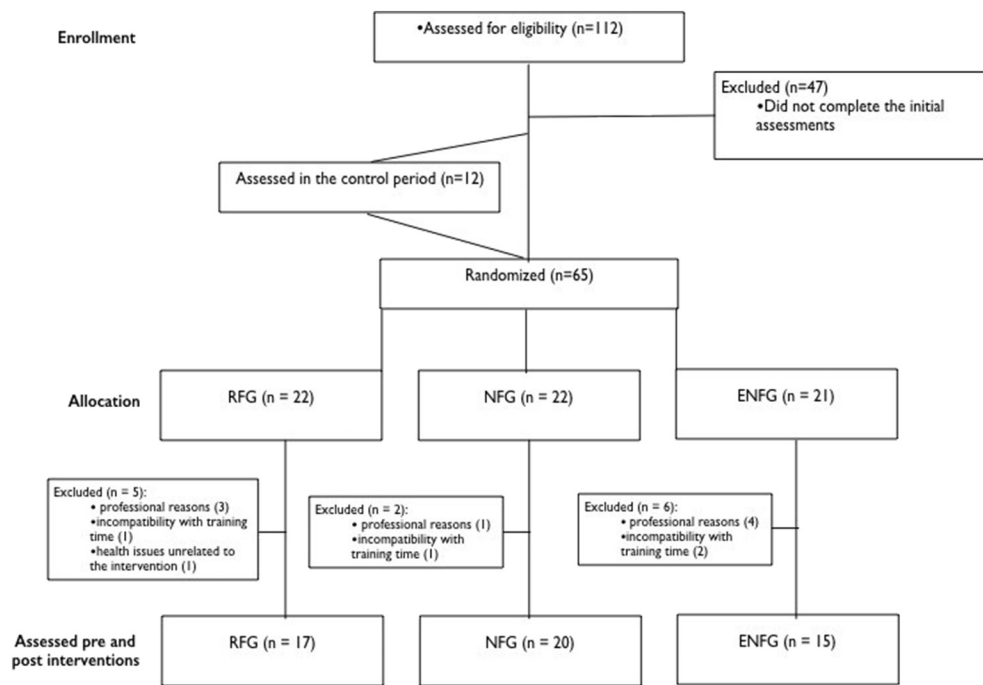


Fig. 1. Flowchart for screening, recruitment, allocation and intervention. RFG, repetitions to failure group; NFG, repetitions not to failure group; ENFG, repetitions not to failure group with equalized volume.

development (RTD) at 50 (RTD₅₀), 100 (RTD₁₀₀), and 250 ms (RTD₂₅₀) ($N \times s^{-1}$). The RTD variables were calculated from the force onset, which was considered the point that the force exceeded 2.5% of the peak torque (Aagaard et al., 2002), and were determined using the MATLAB software.

2.5. Maximal neuromuscular activity

During the isometric strength test, the maximal neuromuscular activity of agonist muscles was assessed using surface electromyography (RMS values) in the *vastus lateralis* (VL EMG_{max}) and *rectus femoris* (RF EMG_{max}). Electrodes were positioned on the muscular belly in a bipolar configuration (20 mm interelectrode distance) in parallel with the orientation of the muscle fibers, according to Leis and Trapani (2000). Shaving and abrasion with alcohol were carried out on the muscular belly in order to maintain the interelectrode resistance above of 2000 Ω . The electrode position was carefully mapped using a transparent paper to ensure identical positioning for pre- and post-testing (Narici et al., 1989). The ground electrode was fixed on the anterior crest of the tibia. The raw EMG signal was acquired simultaneously with the MVC using a four-channel electromyograph (Miotoool, Miotec, Porto Alegre, Brasil). The raw EMG was converted by an A/D converter Miograph with 16 bits resolution (Miotec, Porto Alegre, Brasil), with a sampling frequency of 2000 Hz per channel, connected to a computer. Following the

acquisition of the signal, the data were exported to the MATLAB software, where they were filtered using the Butterworth band-pass filter of 4th order, with a cut-off frequency between 20 and 500 Hz. After that, the EMG records were sliced exactly in the second when maximal value of stable force was determined between the 2nd and 4th second of the force-time curve, and the root mean square (RMS) values were calculated.

2.6. Muscle thickness

The muscle thickness (MT) was measured using B-mode ultrasound (Nemio XG, Toshiba, Japan). A 9.0-MHz linear array probe (38 mm) was placed on the skin perpendicular to the tissue interface, and the scanning head was coated with a water-soluble transmission gel to provide acoustic contact without depressing the dermal surface. The images were digitalized and after analyzed in software Image-J (National institute of health, USA, version 1.42). The subcutaneous adipose tissue-muscle interface and the muscle-bone interface were identified, and the distance from the adipose tissue-muscle interface was defined as MT. The MT images were determined in the muscles *vastus lateralis* (VL), *vastus medialis* (VM), *vastus intermedius* (VI) and *rectus femoris* (RF). Positioning to the measurement for each muscle is described elsewhere (Cadore et al., 2013), and in the Supplementary file. The sum of the four lower-body muscles MT was considered as

Table 1
Physical characteristics pre- and post-interventions (Mean $\pm$ SD).

	RFG		NFG		ENFG	
	Pre	Post	Pre	Post	Pre	Post
Age (years)	66.1 $\pm$ 5.0	66.2 $\pm$ 5.1	66.7 $\pm$ 6.1	67.0 $\pm$ 6.1	65.6 $\pm$ 3.4	66.1 $\pm$ 3.2
Height (cm)	171 $\pm$ 8	171 $\pm$ 8	170 $\pm$ 6	170 $\pm$ 6	172 $\pm$ 7	172 $\pm$ 7
Weight (kg)	79.4 $\pm$ 10.6	79.2 $\pm$ 11.0	80.3 $\pm$ 10.6	80.4 $\pm$ 10.9	87.9 $\pm$ 0.1	88.2 $\pm$ 13.8
BMI (kg/cm ²)	27.1 $\pm$ 3.2	26.9 $\pm$ 3.1	27.7 $\pm$ 2.8	27.7 $\pm$ 2.9	30.9 $\pm$ 4.8	30.3 $\pm$ 4.5
Body fat (%)	24.2 $\pm$ 5.4	23.1 $\pm$ 5.3*	24.4 $\pm$ 5.0	23.6 $\pm$ 4.9*	27.3 $\pm$ 6.7	25.9 $\pm$ 6.1*

RFG, repetitions to failure group; NFG, repetitions not to failure group; ENFG, repetitions not to failure group with equalized volume; BMI, body mass index.

* Significant different from pre training values: $P < 0.001$.

Table 2
Strength training volume (sets and repetitions) during 12 weeks of different interventions.

Leg press exercise - weeks		1	2	3	4	5	6	7	8	9	10	11	12
RFG	Reps	16.1 ± 4.6	19.8 ± 5.9	20.2 ± 5.9	22.2 ± 7.6	19.1 ± 5.7	19.7 ± 5.4	20.0 ± 6.6	20.4 ± 8.9	18.0 ± 6.4	17.7 ± 6.8	19.3 ± 6.7	11.3 ± 4.2
	Mean ± SD												
	95% CI	15.0–17.2	18.4–21.3	18.8–21.6	20.3–24.0	17.8–20.5	18.3–21.0	18.4–21.5	18.2–22.5	16.8–19.3	16.4–19.0	18.0–20.6	10.3–12.3
NFG	Sets	2	2	2	2	2	2	2	2	3	3	3	2
	Reps	8	8	9	9	8	8	9	10	8	8	9	7
	95% CI												
ENFG	Sets	2	2	2	2	2	2	2	2	3	3	3	2
	Reps	8	8	9	9	8	8	9	10	8	8	9	7
	95% CI												
Knee extension exercise - weeks	Reps	11.5 ± 3.5	13.4 ± 3.6	14.0 ± 3.5	14.3 ± 4	12.1 ± 3.6	12.7 ± 3.2	12.4 ± 2.8	12.6 ± 3.6	11.0 ± 2.9	10.7 ± 3.9	11.0 ± 3.2	7.7 ± 2.9
	Mean ± SD												
	95% CI	10.7–12.4	12.6–14.3	13.2–14.8	13.3–15.2	11.2–12.9	11.9–13.5	11.7–13.0	11.7–13.4	10.4–11.5	9.9–11.4	10.4–11.6	7.0–8.5
RFG	Sets	2	2	2	2	2	2	2	2	3	3	3	2
	Reps	6	6	6	6	5	6	6	6	6	6	6	4
	95% CI												
NFG	Sets	2	2	2	2	2	2	2	2	3	3	3	2
	Reps	6	6	6	6	5	6	6	6	6	6	6	4
	95% CI												
ENFG	Sets	2	2	2	2	2	2	2	2	3	3	3	2
	Reps	6	6	6	6	5	6	6	6	6	6	6	4
	95% CI												

RFG, repetitions to failure group; NFG, repetitions not to failure group; ENFG, repetitions not to failure group with equalized volume; Reps, repetitions; 95% CI, 95% interval confidence. Note: repetitions from RFG group are presented in mean, SD and 95% CI because this group performed repetitions until concentric failure and the number ranged among different individuals.

representative of quadriceps femoris (QF) muscle thickness (QF MT). The MT values was considered the mean of three different images taken pre- and post-training. To ensure the same probe position in subsequent tests, the right thigh of each subject was mapped for the position of the electrodes moles and small angiomas by marking on transparent paper (Cadore et al., 2013). Participants were evaluated in a supine position, after 15 min resting and after 72 h without any vigorous physical activity. The MT test-retest reliability coefficients (ICCs) were 0.96 for VL, 0.93 for VM, 0.94 for VI and 0.95 for RF. The calculated typical error was 0.77 mm for VL, 1.36 mm for VM, 1.12 mm for VI, and 0.84 mm for RF.

2.7. Specific tension

Specific tension, a parameter to assess muscle quality, was calculated from the quotient between the maximal dynamic strength (1RM) of the right leg and the sum of the muscle thickness (MT) of the muscles of quadriceps femoris (QF). Thus, specific tension (ST) was calculated following the formula: $ST (kg \cdot mm^{-1}) = 1RM (kg) \text{ of the right leg} / QFMT (VL + VM + VI + RF) (mm)$ (Pinto et al., 2014).

2.8. Jump performance

The jump performance was measured by highest height reached in Squat Jump (SJ) and Counter Movement Jump (CMJ). On the previous day, participants were familiarized with evaluation, jump performance, rest time and criteria's for considering the valid jump. Participants were positioned above the platform of jumps (Cefise, São Paulo, Brazil), and oriented to perform the jumps with hands on the hips and maintain the knees and hips extended during flight phase. Jump height was determined using an acknowledged flight-time calculation (Bosco and Rusko, 1983) in the software Jump System Pro 1.0. Description of individuals' initial positioning, and execution of SJ and CMJ tests are described in the Supplementary file. Participants performed five attempts for each jump, with 20 s of rest between attempts. A variation of 3% was established as minimum between valid attempts. For analysis was considered the highest valid jump.

2.9. Peak oxygen uptake

In order to determine the intensity of endurance training, participants performed an incremental test on a treadmill (Cybex, USA), in order to determine the peak oxygen uptake (VO_{2peak}). The modified Bruce protocol, with 3 min stages was used. All the incremental tests were conducted in the presence of a physician. The breath-by-breath expired gas was analyzed using a metabolic cart (Quark CPET, Cosmed, Italy). The maximum VO_2 value ($ml \cdot kg^{-1} \cdot min^{-1}$) obtained close to exhaustion was considered the VO_{2peak} . More details regarding test's procedures are described in the Supplementary file. The heart rate (HR) was measured using a heart rate monitor (Polar model 2610, Finland). The ICC value was 0.88 for VO_{2peak} .

2.10. Training interventions

Participants of the study trained both strength and endurance training in the same session, twice weekly, on non-consecutive days, and all groups always performed strength prior to endurance training. This intra-session exercise order (i.e., strength prior to endurance training) was chosen in order to optimize neuromuscular gains (Cadore et al., 2013). Importantly, the differences between training interventions occurred only in the exercises to quadriceps muscles (i.e., bilateral leg press and bilateral knee extension), because we focused the investigation on those muscles. Therefore, during the intervention period, RFG performed concurrent training with the ST workout performed using repetitions until concentric failure in all sets in the target exercises; NFG performed concurrent training with the ST workout

Table 3

Control period data (Mean $\pm$ SD) (n = 12), test-retest reliability coefficient (ICCs) and typical error of measurement.

	Week -4	Week 0	Typical error	ICC values
LP 1RM (kg)	145 $\pm$ 75	150 $\pm$ 68	6.82	0.99
KE 1RM (kg)	72 $\pm$ 20	75 $\pm$ 22	1.37	0.95
QF MT (mm)	80.1 $\pm$ 11.4	79.4 $\pm$ 13.4	1.9	0.99
RTD ₅₀ (N·m ⁻¹)	520 $\pm$ 232	505 $\pm$ 192	21	0.88
RTD ₁₀₀ (N·m ⁻¹)	525 $\pm$ 249	504 $\pm$ 203	24	0.91
RTD ₂₅₀ (N·m ⁻¹)	390 $\pm$ 177	377 $\pm$ 149	22	0.90
PT _{iso} (N·m)	197 $\pm$ 34	192 $\pm$ 40	13.4	0.88
SJ (cm)	14.0 $\pm$ 6.2	14.5 $\pm$ 6.3	1.1	0.95
CMJ (cm)	14.8 $\pm$ 5.2	15.1 $\pm$ 6.0	0.76	0.95
VL EMG _{max} (MV)	213 $\pm$ 116	198 $\pm$ 100	19.1	0.83
RF EMG _{max} (MV)	158 $\pm$ 83	160 $\pm$ 69	14.0	0.90

LP 1RM and KE 1RM, maximal dynamic strength in the leg press and knee extension exercises, respectively; MT, muscle thickness; VL, *vastus lateralis*; VM, *vastus medialis*; VI, *vastus intermedius*; RF, *rectus femoris*; QF, *quadriceps femoris*; RTD₅₀, RTD₁₀₀ and RTD₂₅₀, rate of torque development at 50, 100 and 250 ms, respectively; PT_{iso}, isometric peak torque; SJ, squat jump; CMJ, counter-movement jump; EMG_{max}, maximal neuromuscular activity. No significant differences during control period. Note: typical error values are presented in the specific unit of each measurement.

performed using 50% of the possible repetitions maximal (RM) in the target exercises; and, ENFG performed concurrent training with the ST workout performed using 50% of the possible RM in the target exercises, but with the doubling of sets to equalize the total ST volume with RFG in these exercises. Familiarization with exercises and warm-up are described in details in the Supplementary file. The number of sets and repetitions during the intervention in the leg press and knee extension exercises is shown in the Table 2. From weeks 1 to 4, strength training started at an intensity of 65% of 1 RM. From the week 5 to 8, the intensity was increased to 70% of 1 RM + 5% (i.e., taking into consideration that there was an increase in the 1 RM values, we increase the load of reference by 5%), increasing to 75% of 1 RM + 5% from week 9 to 11. Before the post-training assessments, we provided a tapering week (week 12), reducing the number of sets to 2 sets to each quadriceps exercise, and participants performed 80% of the repetitions performed in the previous week, keeping the training load, in order to reduce a possible residual fatigue in the assessments. In the non-concentric failure groups (NFG and ENFG), the initial number of repetitions was determined in a pilot study testing the average number of repetitions performed by 10 elderly men at different intensities (i.e., 65–80% of 1 RM). We adopted this strategy because participants were randomized, and all groups started the intervention simultaneously. The recovery time between sets was 120 s.

Along with quadriceps exercises, individuals also performed bench press, abdominal exercises, back extension and seated row. In these exercises, participants started performing 2 sets of 18–20 repetitions and progressing loads until 3 sets of 8–10 repetitions per sets. The number of repetitions was approximately 80% of the RM possible with the loads. The endurance training program was performed using a treadmill, at the intensity relative to the maximal heart rate (HR_{max}). During the first two weeks, subjects exercised for 20 min at 60–65% of HR_{max}, progressing to 65–70% of HR_{max} in the weeks 3 and 4. In the weeks 5 and 6, subjects exercised for 25 min at 65–70% of HR_{max}, and progressed to 30 min from the week 7 to 9 at same intensity. In the weeks 10 and 11, subjects trained during 30 min at 70–75% of HR_{max}. As in the ST, we also designed a tapering week (week 12) before the post-training assessment in the endurance training, and subjects performed 25 min at 70–75% of HR_{max}. All training groups performed the same endurance training program. Training sessions were carefully supervised by at least three experienced personal trainers.

2.11. Statistical analysis

The SPSS statistical software package was used to analyze all data. Normal distribution and homogeneity parameters were checked with Shapiro-Wilk and Levene test's respectively. Results are reported as mean $\pm$ SD. Statistical comparisons in the control period (from week -4 to week 0) were performed by using Student's paired *t*-tests. The training-related effects were assessed using a two-way Analysis of Variance (ANOVA) (group $\times$ time). If a time vs. group interaction was observed, follow up analysis was proceed using one-way ANOVA with Tukey's post hoc test for group factor, and repeated measures ANOVA for time factor. Comparisons between the total volume and training compliance between groups were performed using one-way ANOVA with Tukey's post hoc test. Significance was accepted when $P < 0.05$. The effect size (ES) between pre and post training for each group was calculated using Cohen's *d* ES, represented by the following formula: $ES = (M_{\text{post}} - M_{\text{pre}}) / SD_{\text{pre}}$, which M_{post} is the mean post-training measure, M_{pre} is the mean pre-training measure for each group, and SD_{pre} is the standard deviation of the pre measurements (Nakagawa and Cuthill, 2007). Responsiveness in the maximal strength and quadriceps muscle thickness were calculated considering responders as participants who achieved an increase greater than two times the typical error away from zero (Alvarez et al., 2017). More details on calculation of the effect size and outcomes responsiveness are presented in the Supplementary file.

3. Results

3.1. Screening, recruitment and allocation

Screening, recruitment, allocation are shown in details in the Fig. 1. From one hundred and twelve individuals who volunteered initially to participate in this study, fifty-two elderly men completed the pre- and post-measurements and had their data included in the statistical analysis (RFG: n = 17; NFG: n = 20, ENFG: n = 15).

3.2. Control period, physical characteristics, training compliance, total strength training load and adverse effects

Data on control period, ICCs, and typical error of measurement are presented in the Table 3. Across the control period (i.e., between week -4 and week 0), no significant changes were observed in any variables assessed. Before and after training, there were no differences between groups in body mass (kg), height (cm), age (years) and percent fat (%) (Table 1). There was no difference in the training compliance among three groups (RFG: 99.3 $\pm$ 1.6%; NFG: 99.0 $\pm$ 3.7%; ENFG: 98.0 $\pm$ 3.0%). Regarding total ST load (i.e., sets $\times$ repetitions $\times$ load) (kg), RFG and ENFG presented significant greater total ST load than NFG across the 12 weeks in the LP exercise (114,794 $\pm$ 34,330 kg, 122,952 $\pm$ 33,337 kg, and 52,625 $\pm$ 20,645 kg, for RFG, ENFG and NFG, respectively, $P < 0.001$), as well as in the KE exercise (35,401 $\pm$ 8954 kg, 35,029 $\pm$ 9330 kg, and 16,905 $\pm$ 3411 kg, for RFG, ENFG and NFG, respectively, $P < 0.001$). There were no differences between RFG and ENFG in the total ST load in both LP and KE exercises. Across the intervention period, the participants reported no adverse effects related to training programs.

3.3. Dynamic strength

At baseline, there were no differences between the groups in the leg press (LP) and knee extension (KE) 1 RM (kg). In the LP 1RM, there was a significant time effect ($P < 0.001$), whereas no significant group effect and time vs. group interaction were observed (Fig. 1a). After 12 weeks, all training groups significantly improved the LP 1 RM values (RFG: 45 $\pm$ 18%, ES = 1.56; NFG: 41 $\pm$ 23%, ES = 0.91; ENFG: 45 $\pm$ 19%, ES = 1.64), with no differences demonstrated between the

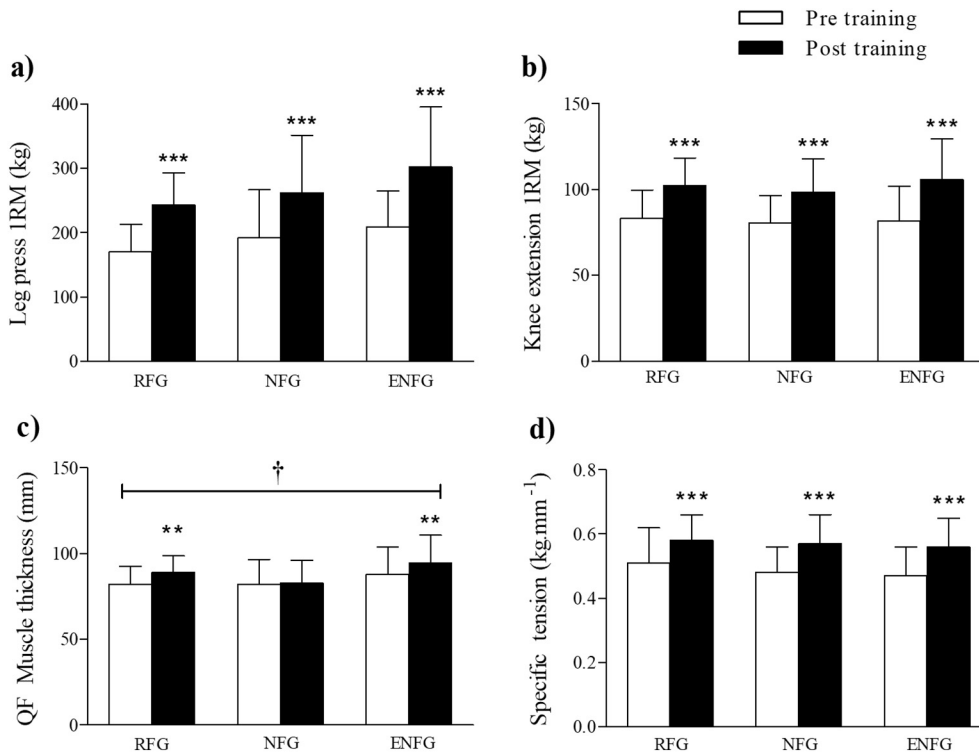


Fig. 2. Mean $\pm$ SD of a) leg press one maximum repetition (1RM) (kg); b) knee extension 1RM (kg); c) quadriceps femoris (QF) muscle thickness (mm); and, d) specific tension values ($\text{kg}\cdot\text{mm}^{-1}$), pre and post 12 weeks of concurrent training. RFG, repetitions to failure group; NFG, repetitions not to failure group; ENFG, repetitions not to failure group with equalized volume. Significant difference from pre training values **($P < 0.01$), ***($P < 0.001$). †Significant time vs. group interaction ($P < 0.05$).

groups (Fig. 2). In the KE 1RM, there was a significant time effect ($P < 0.001$), whereas no group effect and time vs. group interaction were observed (Fig. 1b). After 12 weeks, all training groups significantly improved their KE 1RM values (RFG: $25 \pm 16\%$, ES = 1.12; NFG: $23 \pm 11\%$, ES = 1.06; ENFG: $31 \pm 17\%$, ES = 1.14), with no differences demonstrated between the groups (Fig. 2b). Responders in the LP 1RM were 100% (17/17), 100% (20/20) and 93.33% (14/15) for RFG, NFG and ENFG, respectively. In the KE 1RM, responders were 100% (RFG: 17/17; NFG: 20/20; and ENFG: 15/15) for all three groups.

3.4. Isometric peak torque

At baseline, there were no differences between groups in isometric peak torque (PT_{iso}) (N·m). After training, there was a significant time effect ($P < 0.001$), whereas no group effect and time vs. group interaction were observed. After 12 weeks, all training groups significantly improved their PT_{iso} values (RFG: $11 \pm 9\%$, ES = 0.59; NFG: $10 \pm 12\%$, ES = 0.47; ENFG: $5 \pm 12\%$, ES = 0.26), with no differences between the groups (Table 4).

3.5. Jump performance

At baseline, there were no differences between groups in the SJ and CMJ height (cm). In the SJ, there was a significant time effect ($P < 0.001$), whereas no group effect and time vs. group interaction were observed. After 12 weeks, all training groups significantly improved the SJ performance (RFG: $11.1 \pm 13.1\%$, ES = 0.54; NFG: $4.8 \pm 14.7\%$, ES = 0.21; ENFG: $12.6 \pm 14\%$, ES = 0.67), with no differences between the groups (Table 4). In the CMJ, there was significant time effect ($P < 0.001$), whereas no group effect and time vs. group interaction were observed. After 12 weeks, all training groups significantly improved the CMJ performance (RFG: $7.0 \pm 13.7\%$, ES = 0.33; NFG: $8.5 \pm 14.1\%$, ES = 0.31; ENFG: $7.4 \pm 14.2\%$, ES = 0.43), with no differences between the groups (Table 4).

3.6. Rate of torque development

At baseline, there were no differences between groups in the RTD at intervals of 0–50, 0–100 and 0–250 ms ($\text{N}\cdot\text{m}\cdot\text{s}^{-1}$). In the RTD_{50} , RTD_{100} and RTD_{250} , there were significant time effects ($P < 0.05$, 0.01 and 0.01 for RTD_{50} , RTD_{100} and RTD_{250} , respectively), whereas no group effects and time vs. group interactions were observed. After 12 weeks, all training groups significantly improved their RTD_{50} (RFG: $11 \pm 35\%$, ES = 0.42; NFG: $18 \pm 39\%$, ES = 0.55; ENFG: $11 \pm 36\%$, ES = 0.43), RTD_{100} (RFG: $8 \pm 30\%$, ES = 0.42; NFG: $21 \pm 39\%$, ES = 0.62; ENFG: $11 \pm 32\%$, ES = 0.57) and RTD_{250} (RFG: $8 \pm 17\%$, ES = 0.48; NFG: $17 \pm 28\%$, ES = 0.52; ENFG: $6 \pm 20\%$, ES = 0.38), with no differences between the groups (Table 4).

3.7. Muscle thickness

At baseline, there were no differences between groups in the VL, RF, VI, VM and QF muscle thickness (mm). After training, there was a significant time effect ($P < 0.001$) and time vs. group interaction ($P < 0.05$) in the VL MT. Follow-up analysis showed that there were significant improvements only in RFG ($5.5\% \pm 10.2\%$, $P < 0.05$, ES = 0.26), and ENFG ($9.7 \pm 9.9\%$, $P < 0.001$, ES = 0.38), while no significant changes were observed in NFG ($1.9 \pm 10.8\%$, ES = 0.02) (Table 4). There was a significant time effect in the RF MT ($P < 0.05$), and no significant group effect or time vs. group interaction were observed (RFG: $8.9\% \pm 14.0\%$, ES = 0.42; NFG: $1.2 \pm 8.7\%$, ES = 0.03; ENFG: $4.7 \pm 11.2\%$, ES = 0.10) (Table 4). In the VI MT, there was a significant time effect ($P < 0.05$), no significant group effect, and a trend toward significant time vs. group interaction ($P = 0.1$). Therefore, we decide to proceed with a follow up analysis, which showed that there were significant improvements in VI MT only in RFG ($8.8\% \pm 14.1\%$, $P < 0.05$, ES = 0.31), and ENFG ($9.7 \pm 19.9\%$, $P < 0.05$, ES = 0.51), whereas no significant changes occurred in NFG ($-1.8 \pm 10.7\%$, ES = -0.10 ; ENFG) (Table 4). Regarding VM MT, there was a significant time effect ($P < 0.01$), and no significant group effect or time vs. group interaction were observed (RFG:

Table 4
Neuromuscular performance outcomes and muscle thickness. Mean $\pm$ SD.

	RFG		NFG		ENFG	
	n = 18		n = 21		n = 15	
	Pre	Post	Pre	Post	Pre	Post
LP 1RM (kg)	176 $\pm$ 48	251 $\pm$ 60***	187 $\pm$ 74	255 $\pm$ 86***	208 $\pm$ 57	302 $\pm$ 93***
KE 1RM (kg)	84 $\pm$ 17	103 $\pm$ 16***	80 $\pm$ 16	97 $\pm$ 20***	81 $\pm$ 21	105 $\pm$ 25***
VL MT (mm)	20.8 $\pm$ 3.8	21.8 $\pm$ 3.9*	20.7 $\pm$ 4.3	20.8 $\pm$ 3.6 [†]	21.8 $\pm$ 3.9	23.3 $\pm$ 4.2***
RF MT (mm)	18.3 $\pm$ 3.6	19.8 $\pm$ 3.8*	19.2 $\pm$ 3.7	19.3 $\pm$ 3.5*	19.5 $\pm$ 6.7	20.2 $\pm$ 6.2*
VI MT (mm)	17.3 $\pm$ 4.2	18.6 $\pm$ 3.8*	15.8 $\pm$ 3.8	15.4 $\pm$ 4.1	17.8 $\pm$ 3.2	19.4 $\pm$ 4.5*
VM MT (mm)	25.3 $\pm$ 3.5	27.6 $\pm$ 4.8***	25.6 $\pm$ 6.1	26.2 $\pm$ 5.9***	28.6 $\pm$ 7.8	30.1 $\pm$ 7.1***
QF MT (mm)	81.9 $\pm$ 10.6	89.0 $\pm$ 9.5**	81.9 $\pm$ 14.7	82.5 $\pm$ 13.6 [†]	87.9 $\pm$ 15.8	94.6 $\pm$ 16.1**
Specific tension (kg·mm ⁻¹)	0.51 $\pm$ 0.11	0.58 $\pm$ 0.08***	0.48 $\pm$ 0.08	0.57 $\pm$ 0.09***	0.47 $\pm$ 0.09	0.56 $\pm$ 0.09***
RTD ₅₀ (N·m ⁻¹)	673 $\pm$ 152	737 $\pm$ 270*	528 $\pm$ 153	612 $\pm$ 262*	581 $\pm$ 138	640 $\pm$ 236*
RTD ₁₀₀ (N·m ⁻¹)	669 $\pm$ 136	726 $\pm$ 250**	533 $\pm$ 153	628 $\pm$ 248**	583 $\pm$ 116	649 $\pm$ 214**
RTD ₂₅₀ (N·m ⁻¹)	458 $\pm$ 83	498 $\pm$ 123**	397 $\pm$ 106	453 $\pm$ 133**	438 $\pm$ 76	467 $\pm$ 115**
PT _{iso} (N·m)	197 $\pm$ 34	217 $\pm$ 29***	185 $\pm$ 38	203 $\pm$ 43***	202 $\pm$ 34	211 $\pm$ 36***
SJ (cm)	15.9 $\pm$ 3.3	17.7 $\pm$ 3.4***	16.1 $\pm$ 5.2	17.2 $\pm$ 5.8***	14.5 $\pm$ 4.3	17.4 $\pm$ 5.2***
CMJ (cm)	17.9 $\pm$ 3.6	19.1 $\pm$ 3.6***	16.9 $\pm$ 5.2	18.5 $\pm$ 5.7***	15.8 $\pm$ 4.6	17.8 $\pm$ 5.3***
VL EMG _{max} (MV)	184 $\pm$ 49	211 $\pm$ 69*	166 $\pm$ 50	175 $\pm$ 64*	165 $\pm$ 57	178 $\pm$ 66*
RF EMG _{max} (MV)	153 $\pm$ 69	166 $\pm$ 71	132 $\pm$ 41	180 $\pm$ 173	132 $\pm$ 65	135 $\pm$ 68

RFG, repetitions to failure group; NFG, repetitions not to failure group; ENFG, repetitions not to failure group with equalized volume; LP 1RM and KE 1RM, maximal dynamic strength in the leg press and knee extension exercises, respectively; MT, muscle thickness; VL, *vastus lateralis*; VM, *vastus medialis*; VI, *vastus intermedius*; RF, *rectus femoris*; QF, *quadriceps femoris*; RTD₅₀, RTD₁₀₀ and RTD₂₅₀, rate of force development at 50, 100 and 250 ms, respectively; PT_{iso}, isometric peak torque; SJ, squat jump; CMJ, countermovement jump; EMG_{max}, maximal neuromuscular activity. Significant different from pre training values: *P < 0.05; ***P < 0.01; ***P < 0.001. [†]Significant time vs. group interaction.

10.1% $\pm$ 17.4%, ES = 0.66; NFG: 3.0 $\pm$ 8.7%, ES = 0.10; ENFG: 7.3 $\pm$ 14.0%, ES = 0.20) (Table 4). In the total QF MT, there was a significant time effect (P < 0.001) and a time vs. group interaction (P < 0.05) (Fig. 1c). Follow up analysis showed that there were significant improvements in QF MT only in RFG (9.6% $\pm$ 12.1%, P < 0.01, ES = 0.67) and ENFG (8.2 $\pm$ 10.1%, P < 0.01, ES = 0.42), while no significant changes occurred in NFG (1.1 $\pm$ 6.2%, ES = 0.04) (Fig. 2c). After training, there was a strong trend toward significant differences between NFG and ENFG in the QF MT (P = 0.055). Responders in the QF MT were 58.8% (10/17), 20% (4/20) and 60% (9/15) for RFG, NFG and ENFG, respectively.

3.8. Specific tension

At baseline, there were no differences between groups in specific tension (kg·mm⁻¹). After training, there was a significant time effect (P < 0.001), whereas no group effect and time vs. group interaction were observed. After 12 weeks, all groups significantly improved the specific tension (RFG: 17.0% $\pm$ 20.5%, ES = 0.64; NFG: 19.8 $\pm$ 11.8%, ES = 1.13; ENFG: 20.9 $\pm$ 21.1%, ES = 1.0), with no differences between the groups (Fig. 1d).

3.9. Neuromuscular activity

At baseline, there were no differences between groups in the vastus lateralis (VL) and rectus femoris (RF) neuromuscular activity (EMG_{max}) (mV). In the VL EMG_{max}, there was significant time effect (P < 0.05), whereas no group effect and time vs. group interaction were observed. After 12 weeks, all training groups significantly improved their VL EMG_{max} (RFG: 14% $\pm$ 18%, ES = 0.54; NFG: 7 $\pm$ 31%, ES = 0.17; ENFG: 15 $\pm$ 55%, ES = 0.23), with no differences between groups (Table 4). In the RF EMG_{max}, there was no significant time effect, group effect, and time vs. group interaction.

4. Discussion

The primary finding of the present study was that all training interventions showed similar improvements in maximal dynamic strength, explosive strength, jump performance, isometric peak torque,

and VL neuromuscular activity after 12 weeks in healthy elderly men. This means that performing repetitions until concentric failure does not provide further neuromuscular performance gains. In addition, it also means that even a low number of repetitions relative to the maximum possible (i.e., 50%) optimizes neuromuscular performance gains in elderly men, at least within 12 weeks. Nevertheless, the groups that performed a higher training volume (i.e., RFG and ENFG) showed greater improvements on VL, VI and total QF muscle thickness, which suggest that greater ST volumes may be necessary to optimize muscle hypertrophy in elderly men.

Regarding maximal dynamic (i.e., 1 RM values), and isometric strength (PT_{iso}), the magnitude of improvements observed in the present study is similar to previous studies investigating concurrent training effects in untrained elderly men following similar periods of training (Wood et al., 2001; Izquierdo et al., 2004; Cadore et al., 2010; Cadore et al., 2013). However, our results showed that performing repetitions until concentric failure did not provide additional maximal strength increases, even in comparison to the group that performed 50% of the repetitions per set without equalizing volume. The absence of additional strength increases due to performing repetitions to concentric failure has been previously shown by several studies in young populations (Folland et al., 2002; Izquierdo et al., 2006; Izquierdo-Gabarrén et al., 2010; Sampson and Groeller, 2016; Martorelli et al., 2017). Notwithstanding, our data is novel because, to the best of the authors' knowledge, this is the first study to compare performing repetitions to failure and repetitions not to failure during concurrent training in elderly men. Our results are important because, although repetitions until concentric failure are often used in ST prescriptions in the elderly, inducing marked neuromuscular adaptations (Kraemer et al., 1999; Pinto et al., 2014; Cadore et al., 2012; Radaelli et al., 2014), this method of ST implies longer sets, and consequently, it elicits greater increases in systolic and diastolic blood pressure and heart rate, as well as rate-pressure product (Nery et al., 2010; Lovell et al., 2011; Gjovaag et al., 2016). In addition, repetitions to failure induce greater metabolic impact at the cellular level (i.e., decreases in the ratios ATP/ADP, ATP/AMP, ATP/IMP and pH) and it may require a longer recovery period between sessions (Gorostiaga et al., 2012). Therefore, this type of ST prescription (i.e., repetitions to failure) is not necessary to optimize strength gains in elderly, and could be avoided in order to

reduce the cardiovascular risk, especially in those older adults with hypertension or another cardiovascular disease. Our data agree with previous meta-analyses, which showed that ST intensity, rather than ST volume, explains the improvements on maximal strength (Steib et al., 2010; Peterson et al., 2010; Borde et al., 2015).

Rate of force development (RFD) and muscle power output are strongly associated with the capacity to perform activities of daily living in elderly populations, and strong associations between functional capacity tests performance with muscle power output and RFD have been previously showed in the elderly (Häkkinen et al., 2000; Reid and Fielding, 2012; Casas-Herrero et al., 2013). In addition, although there is a marked decrease in muscle strength, skeletal muscle power and muscle explosiveness decrease at a greater rate than muscle strength with advancing age (Izquierdo et al., 1999a). In the present study, all training groups similarly improved RTD outcomes and jump performances, with no additional benefit of repetitions to failure or greater volumes of repetitions. This finding agrees with previous studies in young populations which showed no additional muscle power output increases, or even a non-responsiveness of subjects performing repetitions to concentric failure (Izquierdo et al., 2006; Izquierdo-Gabarrén et al., 2010; Martorelli et al., 2017).

Surprisingly, marked increases in RTD outcomes and jump performance occurred even when not using an explosive type of ST. However, although we did not provide verbal encouragement to participants to perform the concentric phase as fast as possible, we also did not encourage them to control this phase and perform it slowly. Therefore, participants may have produced a high rate of force development in a large quantity of repetitions during the interventions, especially at the start of movement of lower limb exercises in order to overcome the inertia. Indeed, increases in RTD at different intervals (i.e., 0–50, 0–250 ms) have been shown in healthy elderly performing traditional strength training (Pinto et al., 2014) and concurrent training with no explosive muscle actions during the ST workout (Cadore et al., 2013). Increases in RTD in short intervals (i.e., 0–50 ms) are more related to neural factors, whereas in longer intervals (i.e., 0–250 ms) RTD becomes more strongly influenced by muscle speed-related properties and MVC (Aagaard et al., 2002; Maffiuletti et al., 2016). Thus, the increases in RTD outcomes observed in the present study may be related to adaptations in maximal neuromuscular activity and isometric peak torque. Regarding jump performance, although our interventions did not include jump exercises or exercises using a fast stretching-shortening cycle, all training groups improved markedly these parameters, and these improvements may be related to the RTD and maximal strength adaptations observed, since these variables are also related with SJ and CMJ performances (Villarreal et al., 2009). This result is also very important because jump performance is markedly reduced during aging, and this performance is strongly associated with functional capacity in elderly subjects (Izquierdo et al., 1999). More importantly, our results demonstrate that repetitions until concentric failure are not necessary to promote RTD and jump performance improvements; and, performing 50% of possible repetitions is sufficient stimulus to optimize muscle explosiveness and jump performance.

Our results showed that, even inducing marked neuromuscular performance improvements, low volume of repetitions not to failure, (i.e., NFG) was not able to promote similar muscle hypertrophy compared to ENFG and RFG. While RFG and ENFG showed improvements in MT of all quadriceps muscles, NFG presented only slight changes in RF and VM MT. It is important to highlight that, as in the neuromuscular performance results (i.e., 1 RM, RTD, jump performance), repetitions to concentric failure did not induce additional enhancements in muscle hypertrophy, suggesting that the maximal effort per set does not stimulate further muscle size gains. Although the present study investigates different muscle groups, these results are in agreement with previous studies investigating young populations (Sampson and Groeller, 2016; Martorelli et al., 2017; Nóbrega et al., 2018). Using similar experimental design as the present study, Martorelli et al.

(2017) have shown that only the groups that performed greater ST volumes (i.e., repetitions to concentric failure or not failure equalizing the ST volume by adding more sets) improved elbow flexor muscle thickness in young women. In another study, Sampson and Groeller (2016) did not observe any additional effect of performing repetitions until concentric failure in elbow flexor muscle hypertrophy in young men. The influence of ST total volume on muscle size adaptations has been previously shown in elderly individuals (Radaelli et al., 2014; Borde et al., 2015), whereas no direct evidence of the influence of fatigue during ST sets on muscle hypertrophy has been provided. Moreover, besides demonstrating that it is unnecessary to perform repetitions to concentric failure to optimize muscle hypertrophy, our results also suggest that the improvement on neuromuscular function in healthy elderly men (i.e., maximal and explosive strength) does not necessarily depend on muscle hypertrophy in the short term (i.e., 12 weeks), which also agrees with previous literature (Aagaard et al., 2010).

Increases in the maximal EMG amplitude of the VL muscle were observed in RFG, NFG and ENFG, suggesting that all interventions were efficient in enhancing the maximal voluntary neuromuscular activity. In contrast, no changes were observed in all groups in the EMG_{max} of the RF muscle. The improvement on VL EMG_{max} is in agreement with several studies investigating neural adaptations induced by strength and concurrent training in elderly populations, which suggests greater motor unit recruitment and a higher firing rate among the motor units (Aagaard et al., 2010; Cadore et al., 2010, 2013; Häkkinen et al., 2000). In addition, the observed increase in specific tension also suggests that neural adaptations, along with muscle intrinsic factors (i.e., reduced muscle fat infiltration and improved muscle quality) may explain most of the enhancements in neuromuscular performance (i.e., maximal and explosive strength) observed in the present study (Pinto et al., 2014; Cadore et al., 2012). It has been shown that there is an increase in neuromuscular activity as the duration of the sets increases in relation to the maximum number of possible repetitions, although a plateau in the EMG amplitude is reached before the concentric failure (Sundstrup et al., 2012). Despite acute effects observed in the literature, in the present study, chronic exposure to repetitions to failure did not provide greater improvements in the maximal neuromuscular activity when compared with the groups which performed submaximal repetitions.

Regarding responsiveness to the different interventions, there were a very large number of responders in the maximal dynamic strength (1 RM) variables. These results suggest that, at the ST intensities used in the present study, performing repetitions until concentric failure, as well as performing 100% or 50% of the possible maximal repetitions did not influence the maximal strength responsiveness in healthy elderly men. All interventions were effective in inducing marked strength gains in almost all individuals, which is in agreement with previous studies showing that there are few or no non-responders among elderly individuals, considering maximal strength adaptations induced by ST (Churchward-Venne et al., 2015; Barbalho et al., 2017). When we look at QF MT, although only approximately 60% of participants in the RFG and ENFG were responders, this prevalence was much greater than that observed in NFG, which is in line with the inferential statistical analysis, as well as with the ESs observed, suggesting that greater volumes of ST induce greater responsiveness in muscle size gains. Moreover, considering the results regarding responsiveness in maximal strength and QF MT in NFG, even though an individual is a non-responder to ST in muscle size does not mean that he will be a non-responder in maximal strength. It should also be stated that the use of twice the amount of typical error to define our participants as responders or non-responders may have led to greater incidences of non-responders, since this method is considered as very conservative compared with other methods (Alvarez et al., 2017). As there is increasing interest in so-called “exercise non-responders”; our study provides evidence that dose-responsiveness clearly depends on the outcome of interest.

The present study has strengths and limitations. One could suggest

that the endurance training performed by the participants could have had some influence on the ST adaptations, since they performed concurrent training. However, all training groups performed the same endurance training program, and therefore, a possible influence would be the same among all groups. In addition, when prescribing the endurance training, we avoided the possibility that participants could engage in some endurance activities apart from our intervention, which could have then influenced our results. Moreover, the endurance workout was composed of a low weekly volume and moderate training intensity, and it was always performed after strength training in all sessions, which minimized the possibility of interference effect (Cadore and Izquierdo, 2013). Furthermore, we only assessed the neuromuscular function of the quadriceps muscles, and therefore, it is unclear if the same response would be observed in other muscle groups. Nevertheless, the quadriceps muscle is one of the main muscle groups related to functional capacity in elderly subjects, so our results have important clinical application. Another strength of our study was that this is the first study to verify the influence of performing repetitions until concentric failure or not to failure during concurrent training prescription in elderly men. In addition, along with investigating the effects of performing sets until failure or not to failure, the experimental design of the present study allowed us to also investigate the effects of the volume of repetitions, since one of the groups performed only approximately 50% of the repetitions in comparison to the others.

In summary, concurrent strength and endurance training performed with repetitions to concentric failure did not provide additional gains in the knee extensors maximal strength, explosive strength, muscle hypertrophy, neuromuscular activity, or jump performance in healthy elderly individuals. In addition, even when performing 50% of the possible maximal repetitions with no volume compensation, elderly men optimized neuromuscular performance gains within 12 weeks of training. From a practical standpoint, although repetitions to failure induced marked neuromuscular gains, the performance of submaximal sets promoted the same magnitude of enhancements, showing that this is an efficient alternative for improving neuromuscular function and muscle hypertrophy in elderly individuals. Another practical application of the present study is that greater training ST volumes appear to be necessary in order to optimize muscle hypertrophy in elderly men.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.exger.2018.03.017>.

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